

Anantha Padmanaban Krishna Kumar

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AI Engineer building end-to-end production GenAI systems, from multimodal retrieval and high-speed serving to verified agentic workflows. Ships scalable, real-world applications backed by a deep, practical understanding of model mechanics.

WORK EXPERIENCE

Walmart | AI Engineer

May 2024 – Present

- Shipped a production LangGraph + MCP (Model Context Protocol) multi agent (ReAct planning, dynamic tool selection, self-correcting verification) that resolved a third of escalation-bound queries before they reached a human.
- Architected a high-precision, 5-stage retrieval pipeline on Azure AI Search (RRF fusion, cross-encoder reranking, query rewrite) paired with citation-grounded generation and post-response validation to eliminate hallucinated outputs.
- Trained and deployed a multimodal CLIP retriever (ViT-B/32 + LoRA, InfoNCE on 50M pairs) as candidate generation for Walmart Connect sponsored ads; served on Triton + Milvus HNSW at sub-50ms p99, lifting NDCG@5 by 8%.
- Built a structure-aware 500K-doc ingestion engine with chunk-level diff detection and incremental re-embedding, cutting reindex costs.
- Deployed a 3-tier LLM evaluation and observability suite using a stratified golden set, creating automated deployment gates based on Hit Rate@5 and faithfulness to ensure production reliability.

Tata Elxsi Limited | Machine Learning Engineer

April 2022 – July 2023

- Built an auto-annotation pipeline (SAM + YOLOv8) for image segmentation, reaching 95% annotation accuracy and cutting manual labeling time by 70%, compressing annotation timelines from weeks to days.
- Trained and deployed real-time defect detection and track measurement models for drone-based inspection of a Class I freight rail network, hitting 93-95% accuracy; owned the PyTorch and ONNX pipeline and collaborated with the CUDA team on C++ kernel optimization.
- Containerized and shipped CV pipelines across multiple client projects from POC to production using Docker, AWS SageMaker, S3, and Lambda, with CloudWatch monitoring to standardize delivery and reduce onboarding time for new projects.

SKILLS

Languages & Core	Python, SQL, Bash, PyTorch, Hugging Face, DeepSpeed, ONNX, TensorRT-LLM, Claude Code, Cursor, CUDA.
LLM Stack & APIs	LangChain, LangGraph, LlamaIndex, MCP, vLLM, SGLang, Triton, FastAPI, gRPC, OpenAI, Claude.
Retrieval & Data	Pinecone, Milvus, HNSW, Redis, PostgreSQL, Snowflake, Databricks.
LLMOps & Eval	Ragas, DeepEval, LangSmith, Langfuse, Guardrails AI, MLflow, W&B, Prometheus, Grafana.
Cloud & DevOps	AWS, GCP, Azure, Docker, Kubernetes, Terraform, Airflow, GitHub Actions, Git.

PROJECTS

LYNX [[GitHub](#)]

- Built and deployed a full-stack AI chat product (Next.js, PostgreSQL/Cloud SQL, Cloud Run): streaming Gemini responses, Langfuse observability and hosted prompt management, and a multimodal-ready JSONB message store with parent-id branching.
- Added Auth0 authentication (JWKS JWT verification), crash-safe streaming persistence with a durable message lifecycle, and transactional denormalized counters powering the conversation list.

CrossModal RAG [[GitHub](#)]

- Built a multimodal RAG (Retrieval-Augmented Generation) assistant over text and images (Qwen3-VL-4B via FastAPI + vLLM, async pooling, structured output parsing), with voice input via Whisper Large + Silero VAD hitting sub-2s voice-to-response latency.
- Engineered an agentic layer (LangGraph, MCP, Amazon Bedrock/Claude) orchestrating multi-step retrieval and tool-driven vector store ops, routing between retrieval, tool calls, and direct answers with structured abstention to minimize hallucinated citations.

EffiLLaMA [[GitHub](#)]

- Shipped a LLaMA 3.2 8B-Instruct assistant onto iPhone by compressing the model from 16GB to 4.5GB via GPTQ quantization while preserving sub-second median token latency, working within the device's hard memory budget.
- Fine-tuned LLaMA 3-8B under memory-constrained training with DeepSpeed ZeRO-3 + LoRA/QLoRA, reducing perplexity by 9% on the eval set; configured tensor parallelism and CPU/GPU offloading to fit training on available hardware.

SELECTED PUBLICATIONS

- Anantha PKK.** *Mechanisms of Non-Monotonic Scaling in Vision Transformers.* [[arXiv](#)] [[GitHub](#)]
- Anantha PKK.** *Semantic Anchors in ICL: Why Small LLMs Can't Flip Their Labels.* [[arXiv](#)] [[GitHub](#)]
- Anantha PKK.** *Parameter Reduction in ViT: A Comparative Study of Sharing and Pruning.* [[arXiv](#)] [[GitHub](#)]
- Z Tian, **Anantha PKK**, H Krishnakumar, R Rawassizadeh. *Attentions Under the Microscope: A Comparative Study.* [[arXiv](#)] [[GitHub](#)]

EDUCATION

Boston University | Master of Science - Data Science
Sastra University | Bachelor of Technology - Computer Science

January 2025
May 2022